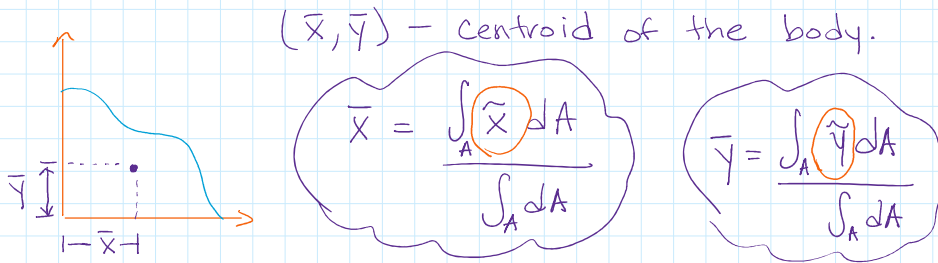


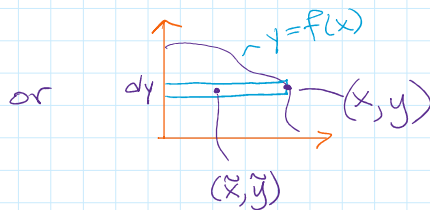
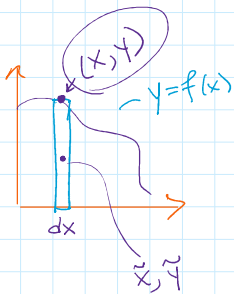
9. Center of Gravity and Centroid

Thursday, October 31, 2024 9:38 AM

- * Center of Gravity - the point on body where you place the weight vector.
- * Center of Mass - same as center of gravity if the body is under a uniform gravitational field.
- * Centroid of a Volume - same as center of mass for a homogeneous material, where ρ - constant.
- * Through geometry, we'll solve for centroids of various bodies.



$\tilde{x}$ - replaced with $f(x)$ or $f(y)$
 $\tilde{y}$ - replaced with $f(x)$ or $f(y)$
 dA - replaced with some product with dx or dy .



$$dA = y dx$$

$$\tilde{x} = x$$

$$\tilde{y} = \frac{y}{2}$$

$$dA = x dy$$

$$\tilde{x} = \frac{x}{2}$$

$$\tilde{y} = y$$

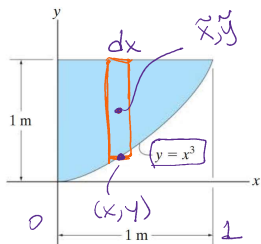
Centroid of Line (slender pipe or rod)

$$\bar{x} = \frac{\int_L \tilde{x} dL}{\int_L dL}, \quad dL = \left[\left(\frac{dx}{dy} \right)^2 + 1 \right]^{1/2} dy$$

$$\bar{y} = \frac{\int_L \tilde{y} dL}{\int_L dL}, \quad dL = \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{1/2} dx$$

$$y = \frac{\int_1 y dL}{\int_1 dL} \quad , \quad dL = \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{1/2} dx$$

F9-1. Determine the centroid $(\bar{x}, \bar{y})$ of the shaded area.



Prob. F9-1

$$\tilde{x} = x$$

$$\tilde{y} = \frac{y+1}{2} = \frac{x^3+1}{2}$$

$$dA = (1-y) dx$$

$$\hookrightarrow dA = (1-x^3) dx$$

$$\tilde{y} = y + h$$

$$\tilde{y} = y + \frac{1-y}{2} = y + \frac{1}{2} - \frac{y}{2}$$

$$= \frac{y}{2} + \frac{1}{2} = \frac{y+1}{2}$$

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA}$$

$$A = \int_0^1 (1-x^3) dx$$

$$A = x - \frac{x^4}{4} \Big|_0^1 = 1 - \frac{1^4}{4} = 1 - \frac{1}{4}$$

$$A = \frac{3}{4} = 0.75 \quad \leftarrow \text{denominator}$$

$$\bar{x} = \frac{\int_0^1 \tilde{x} (1-x^3) dx}{0.75} = \frac{\int_0^1 x - x^4 dx}{0.75} = \frac{\frac{x^2}{2} - \frac{x^5}{5} \Big|_0^1}{0.75}$$

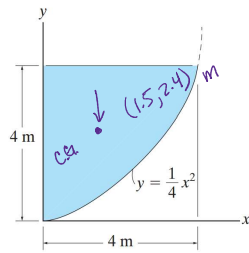
$$= \frac{\frac{1^2}{2} - \frac{1^5}{5}}{0.75} = \frac{\frac{1}{2} - \frac{1}{5}}{0.75} = \frac{0.3}{0.75} = \frac{3}{10} \times \frac{4}{3} = 0.4 \text{ m}$$

$$\bar{y} = \frac{\int_0^1 \tilde{y} (1-x^3) dx}{0.75} = \frac{\int_0^1 \left(\frac{x^3+1}{2} \right) (1-x^3) dx}{0.75} = \frac{\int_0^1 \left(\frac{x^3}{2} - \frac{x^6}{2} + \frac{1}{2} - \frac{x^3}{2} \right) dx}{0.75}$$

$$= \frac{\int_0^1 \left(\frac{1}{2} - \frac{x^6}{2} \right) dx}{0.75} = \frac{\frac{1}{2}x - \frac{x^7}{14} \Big|_0^1}{0.75} = \frac{\frac{1}{2} - \frac{1}{14}}{0.75}$$

$$= \frac{\frac{1}{2} - \frac{1}{14}}{0.75} = \frac{\frac{26}{14} \times \frac{4}{3}}{0.75} = \frac{8}{14} = \frac{4}{7} = 0.57 \text{ m}$$

9-10. Locate the centroid $\bar{y}$ of the shaded area.



Probs. 9-9/10

$(\bar{x}, \bar{y})$ -

$$\bar{x} = \frac{\int \tilde{x} dA}{\int dA} \quad \bar{y} = \frac{\int \tilde{y} dA}{\int dA}$$

$f(x)$ or $f(y)$, $f(dx)$, $f(dy)$

$$y = \frac{1}{4}x^2$$

$$\tilde{x} = \frac{x}{2}$$

$$\tilde{y} = y$$

$$dA = x dy$$

$$x = (4y)^{1/2}$$

denominator: $\int dA = \int_0^4 x dy = \int_0^4 (4y)^{1/2} dy$

$$A = \int_0^4 2y^{1/2} dy = 2 \cdot \frac{2}{3} y^{3/2} \Big|_0^4 = \frac{4}{3} y^{3/2} \Big|_0^4 = \frac{4}{3} (4)^{3/2} - 0$$

$$A = \frac{32}{3} = 10\frac{2}{3} = 10.67 \text{ m}^2$$

$$\bar{x} = \frac{\int \tilde{x} dA}{A} = \frac{\int_0^4 \frac{x}{2} \cdot x dy}{A} = \frac{\int_0^4 \frac{x^2}{2} dy}{A} = \frac{\int_0^4 \frac{4y}{2} dy}{A}$$

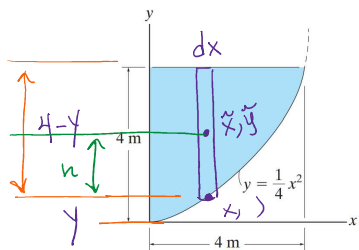
$$\bar{x} = \frac{y^2 \Big|_0^4}{10.67} = \frac{16 - 0}{10.67} = \frac{16}{\frac{32}{3}} = 16 \cdot \frac{3}{32} = 1.5 \text{ m}$$

$$\bar{y} = \frac{\int \tilde{y} dA}{A} = \frac{\int_0^4 y \cdot x dy}{A} = \frac{\int_0^4 y \cdot 2y^{1/2} dy}{A} = \frac{\int_0^4 2y^{3/2} dy}{A}$$

$$\bar{y} = \frac{2 \cdot \frac{2}{5} y^{5/2} \Big|_0^4}{A} = \frac{\frac{4}{5} (4)^{5/2} - 0}{\frac{32}{3}} = \frac{\frac{4}{5} \cdot 32}{\frac{32}{3}} = \frac{12}{5}$$

$$\bar{y} = 2.4 \text{ m}$$

9-10. Locate the centroid $\bar{y}$ of the shaded area.



Probs. 9-9/10

$$\tilde{y} = \frac{4-y}{2} + y = 2 - \frac{y}{2} + y$$

$$\tilde{x} = x$$

$$\tilde{y} = y + h$$

$$\tilde{y} = \frac{y}{2} + 2$$

$$dA = (4-y) dx$$

$$\bar{X} = \frac{\int_0^4 x (4-y) dx}{A} = \frac{\int_0^4 x (4 - \frac{1}{4}x^2) dx}{A}$$

$$\bar{X} = \frac{\int_0^4 4x - \frac{1}{4}x^3 dx}{A}$$

$$\bar{X} = \frac{2x^2 - \frac{1}{16}x^4 \Big|_0^4}{A} = \frac{2(4)^2 - \frac{1}{16}(4)^4 - 0}{\frac{32}{3}}$$

$$\bar{X} = \frac{32 - 16}{32/3} = 16 \cdot \frac{3}{32} = 1.5 \text{ m}$$

$$\bar{y} = \frac{\int_0^4 (\frac{y}{2} + 2) (4-y) dx}{A} = \frac{\int_0^4 (\frac{1}{8}x^2 + 2) (4 - \frac{1}{4}x^2) dx}{A}$$

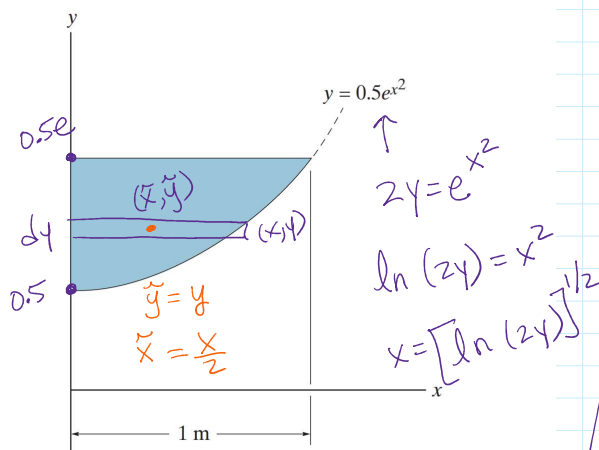
$$\int_0^4 \frac{1}{2}x^2 - \frac{1}{32}x^4 + 8 - \frac{1}{2}x^2 dx = \int_0^4 8 - \frac{1}{32}x^4 dx$$

$$\frac{-\frac{1}{160}x^5 + 8x \Big|_0^4}{32/3} = \frac{-\frac{1}{160}(4)^5 + 8(4)}{32/3} = \frac{32 - \frac{1024}{160}}{32/3}$$

$$\frac{32 - 32/5}{32/3} = \frac{1 - \frac{1}{5}}{1/3} = \frac{4}{5} \cdot \frac{3}{1} = \frac{12}{5} = 2.4 \text{ m}$$

9-15. Locate the centroid $\bar{x}$ of the shaded area. Solve the problem by evaluating the integrals using Simpson's rule.

*9-16. Locate the centroid $\bar{y}$ of the shaded area. Solve the problem by evaluating the integrals using Simpson's rule.



Probs. 9-15/16

$$f(0) = 0.5e^{0^2} = 0.5$$

$$f(1) = 0.5e^{1^2} = 0.5e$$

$$A = \int dA = \int_{0.5}^{0.5e} [\ln(2y)]^{1/2} dy$$

Simpson's Rule

$$\frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

$$f(y) = [\ln(2y)]^{1/2}$$

$$a = 0.5$$

$$b = 0.5e$$

$$\frac{0.5e - 0.5}{6} \left[[\ln(2 \cdot 0.5)]^{1/2} + 4 \left[\ln\left(2 \cdot \left(\frac{0.5 + 0.5e}{2}\right)\right) \right]^{1/2} + [\ln(2 \cdot 0.5e)]^{1/2} \right]$$

$$A = 0.143 \left[0 + 4(0.787) + 1 \right] = 0.593 \leftarrow A$$

$$\bar{X} = \int_{0.5}^{0.5e} \frac{\ln(2y)}{2} dy, \quad a = 0.5, \quad b = 0.5e, \quad f(y) = \frac{1}{2} \ln(2y)$$

$$\frac{0.5e - 0.5}{6} \left[\frac{1}{2} \ln(2 \cdot 0.5) + \frac{4}{2} \ln \left(2 \cdot \left(\frac{0.5 + 0.5e}{2} \right) \right) + \frac{1}{2} \ln(2 \cdot 0.5e) \right]$$

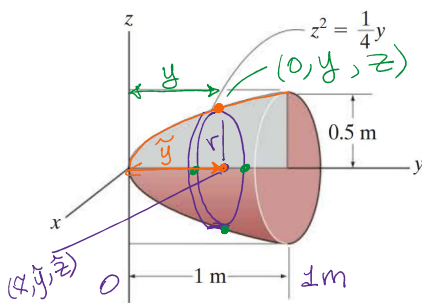
$$0.143 [0 + 1.24 + 0.5] = 0.249$$

$$\bar{x} = \frac{0.249}{0.593} = 0.42 \text{ m}$$

$$\frac{\text{Actual Answer}}{0.3982}$$

$$5.38\% \text{ error}$$

F9-5. Locate the centroid $\bar{y}$ of the homogeneous solid formed by revolving the shaded area about the y axis.



Prob. F9-5

$$\bar{y} = \frac{\int \tilde{y} dV}{\int dV} \quad \text{Centroid of thin disk}$$

$$\tilde{y} = y$$

$$dV = \pi r^2 dy \Rightarrow dV = \pi z^2 dy$$

$$r = z$$

$$r^2 = z^2$$

$$dV = \pi \left(\frac{1}{4} y \right) dy$$

Volume (Denominator)

$$\int dV = \int_0^1 \pi \left(\frac{1}{4} y \right) dy = \frac{\pi}{4} \int_0^1 y dy = \frac{\pi}{4} \left[\frac{y^2}{2} \right]_0^1$$

$$\frac{\pi}{4} \left(\frac{1^2}{2} - 0 \right) = \frac{\pi}{4} \left(\frac{1}{2} \right) = \frac{\pi}{8} \text{ m}^3$$

Numerator

$$\int \tilde{y} dV = \int y dV = \int_0^1 y \left(\frac{\pi}{4} y \right) dy = \frac{\pi}{4} \int_0^1 y^2 dy$$

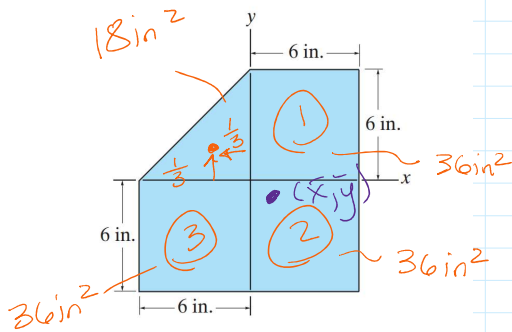
$$\frac{\pi}{4} \cdot \frac{1}{3} y^3 \Big|_0^1 = \frac{\pi}{12} (1^3 - 0) = \frac{\pi}{12} \text{ m}^4$$

$$\frac{\pi}{4} \cdot \frac{1}{3} y^3 \Big|_0^1 = \frac{\pi}{12} (1^3 - 0) = \frac{\pi}{12} \text{ m}^4$$

$$\bar{y} = \frac{\frac{\pi}{12} \text{ m}^4}{\frac{\pi}{8} \text{ m}^3} = \frac{1}{12} \cdot \frac{8}{1} = \frac{2}{3} \text{ m} = 0.67 \text{ m}$$

9-59. Locate the centroid ($\bar{x}$, $\bar{y}$) of the shaded area.

for constant ρ & thickness



Prob. 9-59

$$\bar{x} = \frac{\sum \tilde{x} W}{\sum W} = \frac{\sum \tilde{x} A}{\sum A}$$

$$\bar{y} = \frac{\sum \tilde{y} W}{\sum W} = \frac{\sum \tilde{y} A}{\sum A}$$

	$\tilde{x}$	$\tilde{y}$	$\tilde{x} A$	$\tilde{y} A$
①	+3 in	+3 in	3(36) in ³ 108 in ³	3(36) in ³ 108 in ³
②	+3 in	-3 in	3(36) in ³ 108 in ³	-3(36) in ³ -108 in ³
③	-3 in	-3 in	-3(36) in ³ -108 in ³	-3(36) in ³ -108 in ³
④	-2 in	2 in	-2(18) in ³ -36 in ³	+2(18) in ³ 36 in ³

$$\sum A = 36 + 36 + 36 + 18 = 126 \text{ in}^2$$

$$\sum \tilde{x} A = 72 \text{ in}^3$$

$$\sum \tilde{y} A = -72 \text{ in}^3$$

$$\bar{x} = \frac{\sum \tilde{x} A}{\sum A}$$

$$\bar{y} = \frac{\sum \tilde{y} A}{\sum A}$$

$$\bar{x} = \frac{72 \text{ in}^3}{126 \text{ in}^2} = 0.57 \text{ in} \quad \bar{y} = \frac{-72 \text{ in}^3}{126 \text{ in}^2} = -0.57 \text{ in}$$